In [2]:
```
# Import the following modules on the lines below:
#   numpy, matplotlib.pyplot, and glob
#   You can just add **uncommented** lines to this cell and then evaluate the cell
import numpy as np
import matplotlib.pyplot as plt
import glob
```

In [2]:
```
# Use glob.glob() to read in the names of all the '*.csv' files
for name in glob.glob('*.csv'):
    print(name)
```

```
Teff6875logg5.344.csv
Teff7228logg4.838.csv
Teff7346logg4.987.csv
Teff7364logg4.906.csv
Teff7481logg5.056.csv
Teff7554logg5.012.csv
Teff7594logg4.929.csv
Teff7602logg4.986.csv
Teff7785logg5.040.csv
Teff7811logg5.354.csv
Teff7822logg5.350.csv
Teff7943logg5.611.csv
Teff8197logg5.568.csv
Teff8200logg4.820.csv
Teff8335logg4.938.csv
Teff8557logg5.061.csv
Teff8633logg5.129.csv
Teff8649logg4.824.csv
```

In [12]:
```
# Write code to read in the data in the first file. You can use the
# "loadtxt" command with the appropriate values set for "skiprows" and "delimiter".
# Store the values in the columns labelled "Mag" and "Magerr" in variable names
# of your own choosing.
data = np.loadtxt('Teff6875logg5.344.csv', delimiter=',', skiprows=1)

magnitude = data[:,1]
magnitudeError = data[:,2]
```

In [23]:
```
# Write a function that takes as arguments the list "Mag" and the list "Magerr" fro
# the previous cell. Calculate the standard deviation of "Mag", and divide it by th
# average of "Magerr", and return this value. Test this on the data from the file i
# the cell above to make sure it doesn't cause an error.
def stDevCalc(magnitude, magnitudeError):
    magDev = np.std(magnitude)
    errAvg = np.average(magnitudeError)
    return magDev/errAvg
print (stDevCalc(magnitude, magnitudeError))
```

```
0.5490235630376121
```

In [15]:
```
# To help simplify this lab a little, we provide the function below. If you call it
# with one of the names of the *.csv files, it will pull out the part of the name
# that gives the Teff of the star (as well as its log g value) and return them.
#
```

```
def get_vals(fname):
    Teff = fname[4:8]
    logg = fname[12:17]
    Teff = float(Teff)
    logg = float(logg)
    return Teff,logg

# Example:
f = 'Teff8200logg4.820.csv'
Teff, logg = get_vals(f)
print(Teff,logg)
```

```
8200.0 4.82
```

In [30]:

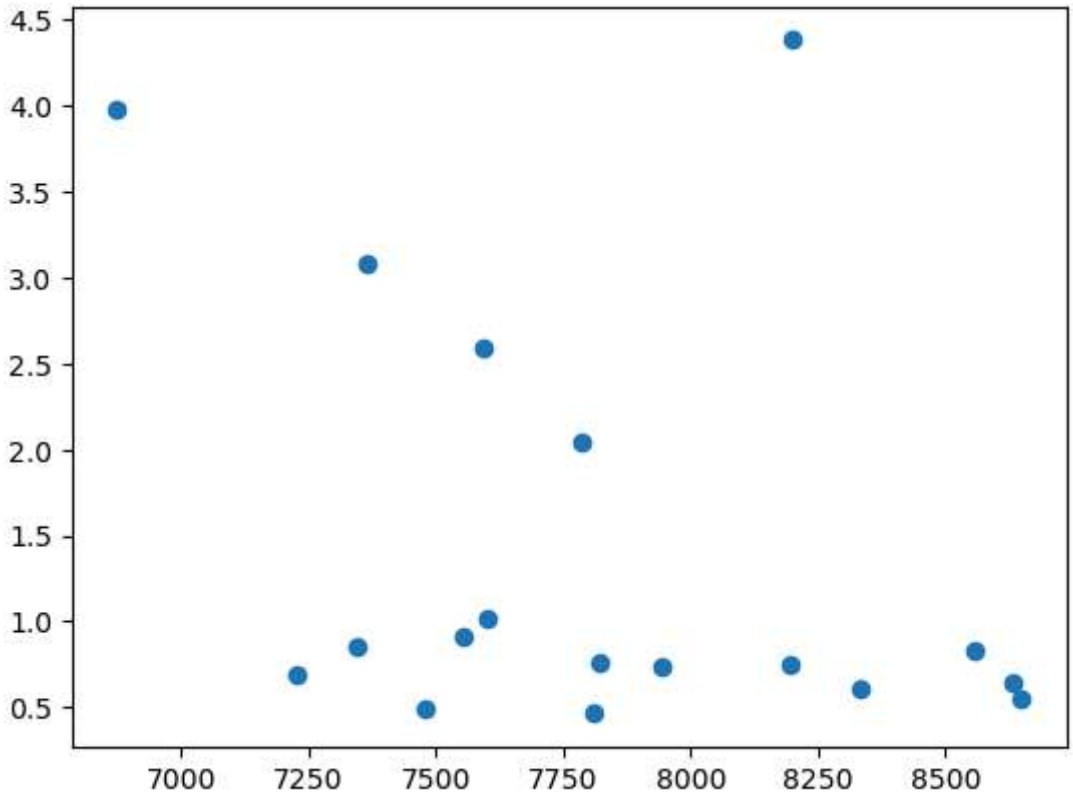
```
# Now, write a for loop that goes through the full list of files. Build up in stage
# the loop does. First, have it print out the name of each file. Then add in the ab
# "get_vals()" command to calculate the temperature and gravity contained in each f
# Now, go back and add in the loadtxt command, so that you read in the list of magn
# the errors on the magnitudes for each star. Finally, call the function that you w
# divides the standard deviation of the magnitudes by "Magerr" to calculate the ext
# that each object possesses.

# We haven't saved any of the values yet, although we have calculated them. Set up
# list for the temperature values and one for the variability values. At the end of
# "append" these the temperature and variability values to their respective lists.

temperature = []
variability = []
for fname in glob.glob('*.csv'):
    temperature.append(get_vals(fname)[0])
    magnitude = np.loadtxt(fname, delimiter=',', skiprows=1)[:,1]
    magnitudeError = np.loadtxt(fname, delimiter=',', skiprows=1)[:,2]
    variability.append(stDevCalc(magnitude, magnitudeError))
```

In [43]:

```
# Make a scatter plot (Google "plt.scatter()") of this: Plot variability (y-axis) v
# Once you have it looking the way you want, save a copy to disk (Google "plt.savef
plt.scatter(temperature, variability)
plt.savefig('lab2.png')
```



In [ ]:
```
# Also, save a PDF copy of your Jupyter Notebook to disk (in the notebook menu do "
# and when the preview shows up do "File > Export as PDF" from the Safari menu **at
# Submit both the PDF of the figure and the notebook on Canvas...
```